CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 10: **The Human Eye and the Colourful World** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A second identical prism is placed upside down in contact with the first one, in the path of a narrow white beam. The emergent beam is 1

(a) only red light **(b)** white light
(c) only violet light **(d)** a wider spectrum

2. A prism of glass is immersed in a liquid whose refractive index equals that of the glass for all colours. A beam of white light passing through it will 1

(a) be reflected **(b)** be absorbed **(c)** go undeviated **(d)** be dispersed

3. Assertion (A): Stars twinkle. Reason (R): Starlight is refracted by varying layers of air. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. A student sees a clear spectrum on a wall when a narrow white beam passes through a prism but sees no spectrum if the beam is replaced by light from a sodium lamp. This is because sodium light 1

(a) is white **(b)** is infrared
(c) is nearly monochromatic **(d)** is invisible

5. A beam of red and violet lights enters a glass slab. In the glass 1

(a) violet travels slower than red **(b)** violet travels faster
(c) both travel at the same speed **(d)** red travels slower than violet

6. Assertion (A): Planets do not twinkle. Reason (R): Planets are much closer than stars and act as extended sources. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. Which of the following is the best way to demonstrate the Tyndall effect? 1

(a) Pass a beam through water with a few drops of milk
(b) Pass a beam through a clean glass block
(c) Pass a beam through pure water
(d) Pass a beam in vacuum

8. Why does a clear sky look white near the horizon? 1

(a) Blue is absorbed
(b) Light of all colours gets scattered many times
(c) No scattering takes place
(d) Only red is scattered

9. **Assertion (A):** Hypermetropia can be corrected using a convex lens. 1
Reason (R): In hypermetropia the eyeball is too long.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The colour of the clear sky at noon is red. 1
Reason (R): Light of shorter wavelength is scattered more strongly than light of longer wavelength.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does a person with hypermetropia not always see distant objects blurred? 2

12. A person's eye lens cannot become thicker than a certain limit. What defect results and why? 2

13. A person has a far point at 60 cm. Find the focal length and power of the corrective lens. 2

14. Explain why twinkling is greater for stars near the horizon. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Why is light of longer wavelength scattered less? Use this to explain the colour of the sky at noon and sunset. 3

16. Explain why the sky is not violet though violet is scattered even more than blue. 3

17. How would the sky appear from the surface of the Moon? Explain. 3

OR

A student looks at a rainbow and sees red on the outer edge and violet on the inner edge. Explain why. 3

18. Why do we see a bright halo of coloured light around a street lamp on a foggy night? 3

OR

Why is the size of the pupil large in dark places and small in bright places? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** In a science class, Kavya allowed a narrow beam of sunlight to fall on a glass prism and obtained a coloured band on a white screen. Her teacher then explained that the same process occurs in the sky after a shower, giving a rainbow. 5
(i) Name the phenomenon by which white light splits into colours.
(ii) Which colour deviates the least and which the most?
(iii) What is the band of colours called?
(iv) Explain how a rainbow is formed in the sky.

20. Describe in steps how rainbow formation depends on 5
(a) refraction
(b) dispersion
(c) total internal reflection
(d) the position of the Sun. Give reasons for the order of colours and the angle.

OR

Explain how the human eye adjusts to a change from bright to dim light and from a distant to a near object. What happens to the pupil, ciliary muscles and focal length in each? 5

END OF QUESTION PAPER — Total: 40 Marks